DRONE MOTOR CONTROL / ENDURANCE AND PAYLOAD

Payload or flight time?

Chiralight makes no compromises.

3 kg quad, 10-inch props, 640KV motor and 6S 4 Ah. 1.5:1 TWR (thrust-to-weight ratio) equivalent flight load; separate 3:1 TWR maneuver reserve for gust recovery and obstacle avoidance.

Design freedom: lower cell counts, lower-KV motors, higher minimum RPM and more maneuvering reserve.

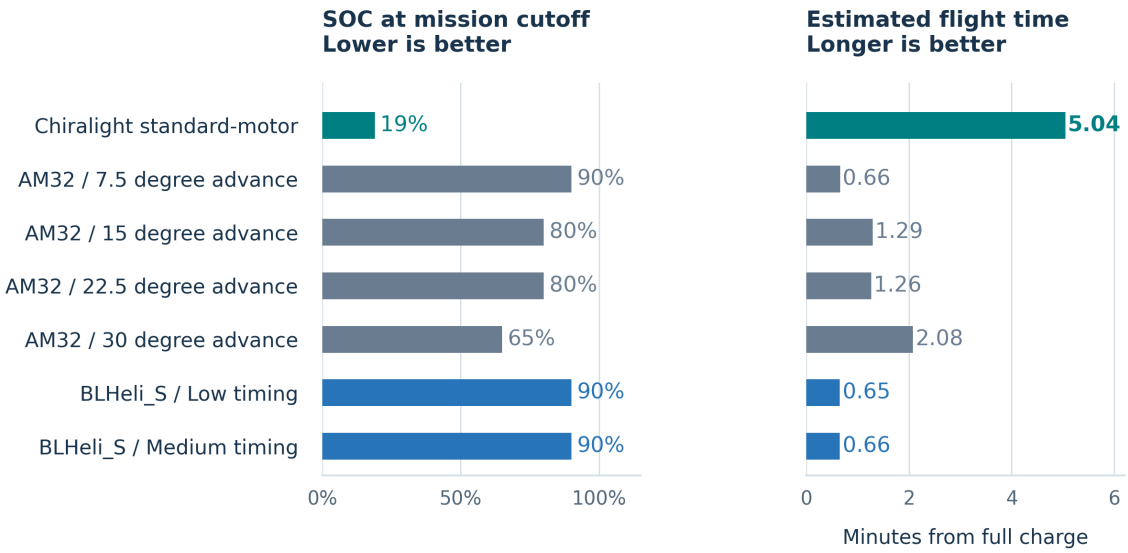
5.0 min
Estimated flight time
Chiralight standard-motor

+143%
Flight time versus the
best tested alternative

+174 g
Potential payload by
choosing 6S over 8S

3:1 TWR MANEUVER RESERVE FOR GUST RECOVERY AND OBSTACLE AVOIDANCE

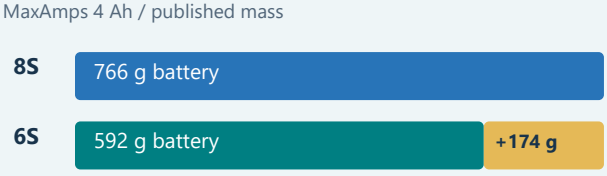
640KV / 6S | 1.5:1 TWR flight load | 3:1 TWR maneuver reserve



AM32 and BLHeli_S show every tested timing setting. The headline compares against **AM32 / 30 degree advance**, the longest-flying tested alternative in this configuration.

6S FOR PAYLOAD. 8S FOR ADDITIONAL VOLTAGE RESERVE.

The smaller 4 Ah pack releases **174 g for payload** at the same 3 kg all-up mass. With Chiralight standard-motor, 8S estimates 7.91 minutes and adds 174 g.



Flight rule: integrate 1.5:1 TWR flight power. Stop at lowest tested SOC meeting 3:1 TWR maneuver reserve for gust recovery and obstacle avoidance, with 3.0 V/cell **MINIMUM** under both loads. Reserve is not continuous. The recommended 3.5 V/cell landing policy is separate.

Model: compiled controllers; nonlinear B/H and table-informed Gemfan 1050 fixture. Synthetic OCV, isothermal 5 milliohms/cell, four motors + 10 W; FOC 1 A injection, MTPA/overmodulation, field weakening off. Equivalent flight estimate, not measured aircraft endurance. Reserve: 125 A mean / 136 A peak against 128 A continuous; no burst rating assumed.

Earlier evidence: +1.9 efficiency points vs AM32; 95% less gust RPM RMS error (p4). 15% lower hover voltage vs AM32 at 7.5 degree advance (p2).

Pack sources: MaxAmps 4 Ah 6S, 592 g / 8S, 766 g. **Verifiable simulation results.** Contact Chiralight Systems. Evaluate for your aircraft. Exact performance may vary.

VOLTAGE AND EFFICIENCY SWEEP

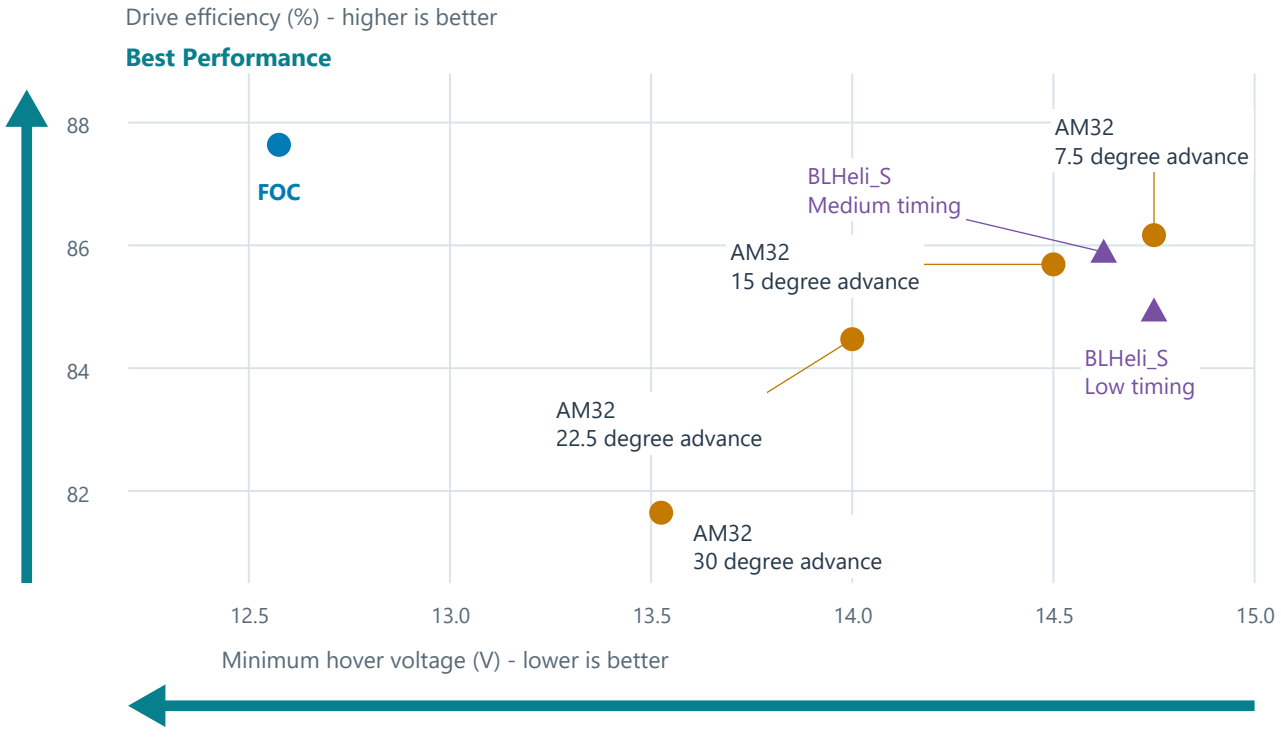
More efficiency. More voltage headroom.

Compiled standard-motor, AM32 and official BLHeli_S firmware share the nonlinear motor and propeller load. FOC: field weakening off, overmodulation on. AM32: fixed advance. BLHeli_S: native adaptive timing.

VERIFIABLE SIMULATION RESULTS / EARLIER VOLTAGE SWEEP

Controller / timing	Drive efficiency	Battery to shafts	Hover voltage
standard-motor FOC	**87.6%**	**83.7%**	**12.6 V**
AM32 / 7.5 degree advance	86.2%	82.2%	14.8 V
AM32 / 15 degree advance	85.7%	81.7%	14.5 V
AM32 / 22.5 degree advance	84.5%	80.5%	14.0 V
AM32 / 30 degree advance	81.6%	77.7%	13.5 V
BLHeli_S / Low timing	84.9%	80.9%	14.8 V
BLHeli_S / Medium timing	85.9%	81.9%	14.6 V

Efficiency: full-charge 6S 5 Ah pack, four motors and 10 W auxiliaries. Battery-to-shaft includes pack losses. Hover voltage: separate test at the same target RPM. Stock-board timing and sensing comparison: page 4.



Standard-motor leads on efficiency and hover voltage. BLHeli_S Medium timing gives 85.9% drive efficiency and needs 14.6 V to hold hover. FOC reaches 87.6% and 12.6 V. AM32 trades about 4.5 efficiency points from 7.5 to 30 degree advance to reach 13.5 V.

Evaluate for your aircraft
Bring your motor, propeller, battery and mission. Contact Chiralight Systems for test details and supporting data.
Exact performance may vary.

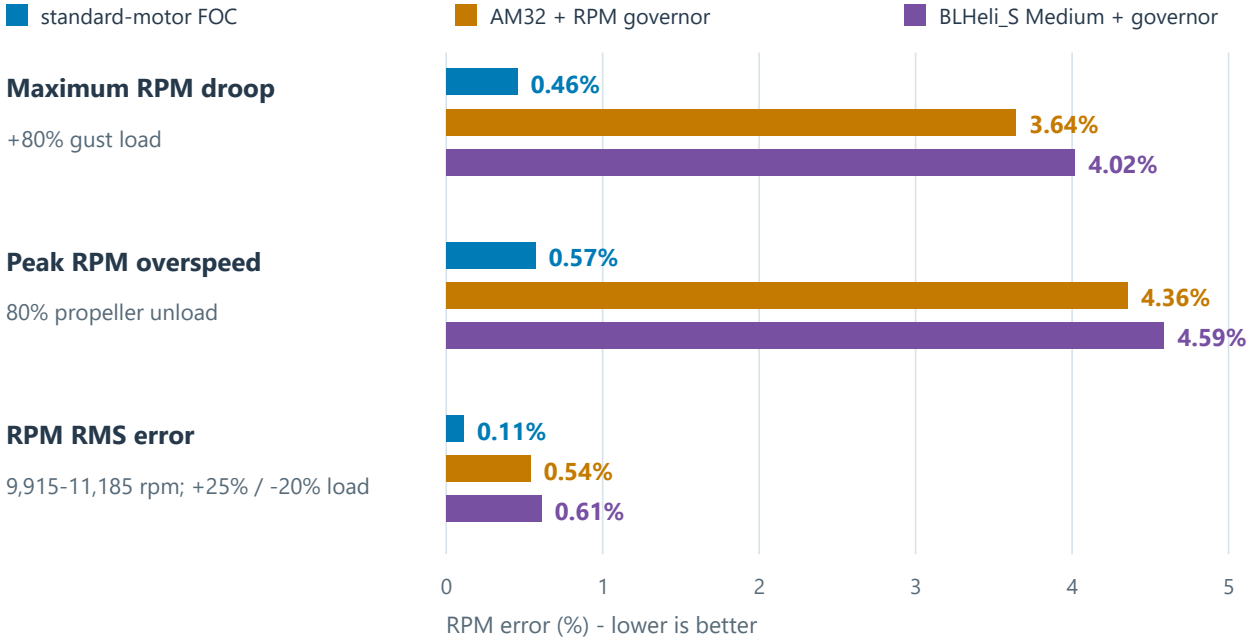
DYNAMIC RESPONSE AND STARTUP

Hold RPM through changing loads.

Earlier controller comparison: speed loops are tuned on separate training runs, then frozen for six dynamic tests. The stock-board profile comparison follows on page 4.

LESS RPM DEVIATION UNDER LOAD

Tested configurations, completed healthy runs; lower is better.



Same nonlinear 24 V motor and propeller load. AM32 and official BLHeli_S Medium use external sensorless RPM governors. No rotor truth enters any controller.

Gust phase RMS current: FOC 25.1 A; AM32 26.5 A; BLHeli_S 26.1 A.

STARTUP AND PROTECTION IN THE TESTED RUNS

FOC: 6 of 6 healthy

All 6 final simulation tests completed without faults or loss of synchronization.

AM32: 2 startup trips

2 of 6 starts tripped after 3 qualification successes. A later braking trip left 3 of 6 runs healthy.

BLHeli_S: 6 starts

All 6 starts succeeded; 4 of 6 runs completed healthy. Two trips during aggressive acceleration.

BLHeli_S dynamic tests use native Medium timing. The separate High-timing fixture tripped in both tested starts; no valid High point is shown on page 2.

AM32 source: maintained upstream am32-firmware/AM32, pinned to 961cb84f5dd4.

Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.

BOARD TIMING AND SENSING / SIMULATION

The advantage survives real inverter timing.

Full 104 ns blanking. Current sensing at 13 ENOB. FOC hover efficiency is about 0.1 percentage point below the ideal timing reference. Actual compiled controllers share the nonlinear B/H motor and propeller load.

+1.9 points
hover efficiency vs AM32

+1.4 points
hover efficiency vs BLHeli_S

95% less
gust RPM RMS error vs both

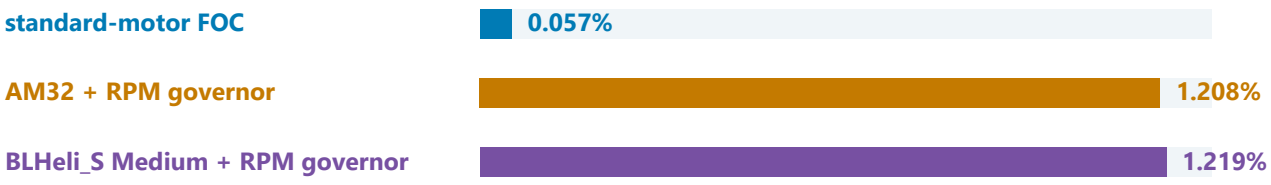
SAME HOVER REQUEST. LOWER ELECTRICAL LOSS.

Controller / electronics	DC-to-shaft	Copper loss	Phase RMS
standard-motor / evaluation-kit 104 ns blanking; 13 ENOB / +/-250 A	**87.52%**	39.52 W	18.15 A
AM32 / SkyStars AM60 E230 416.7 ns blanking; 24 kHz carrier	**85.64%**	44.99 W	19.36 A
BLHeli_S / SpeedyBee BLS 60A J_H_40; 816.3 ns / 23.93 kHz	**86.14%**	43.80 W	19.11 A

Stiff 24 V bus; ~9,915 rpm request; field weakening off. AM32 averages three healthy hover starts. Modeled DC-to-shaft efficiency includes copper and diode losses; unspecified MOSFET and magnetic losses are omitted.

GUSTS: TIGHTER RPM CONTROL

RPM RMS error over the same 1.6-5.0 s gust sequence - lower is better.



Worst gust RPM shortfall: FOC 0.46%; AM32 3.82%; BLHeli_S 4.07%.

Throttle reduction exposes current limits.

FOC limits current and retains lock at a 52.03 A dynamic phase peak; the most aggressive RPM request exceeds available torque. AM32 and BLHeli_S reach the common 100 A bench guard.

Frozen speed-loop design bandwidth: FOC 120 rad/s; AM32 and BLHeli_S 20 rad/s with external RPM governors. All 12 FOC timing/sensing runs retain lock. Stock AM32 hover starts: 3 of 3 healthy.

Verifiable simulation results. Contact Chiralight Systems for test details and supporting data.